NARAYANA IIT ACADEMY INDIA

40+ YEARS OF EXCELLENCE

Sec: SR.IIT_*CO-SC(MODEL-A,B&C) Date:24-01-2024
Time: 3HRS Max. Marks: 300

Name of the Student: ____________________ H.T. NO:

24-01-24_SR.STAR CO-SUPER CHAINA(MODEL-A,B&C)_JEE-MAIN_GTM-26(N)_SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

MATHEMATICS: TOTAL SYLLABUS

SUBJECT	MISTAKES		
	JEE SYLLABUS Q'S	JEE EXTRA SYLLABUS Q'S	TOTAL Q'S
MATHS			
PHYISCS			
CHEMISTRY			

THE NARAYANA GROUP

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

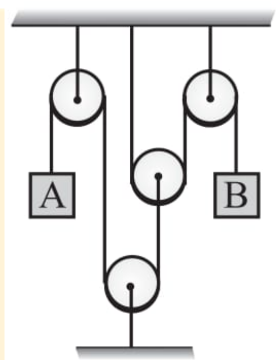
1. What are the dimensions of A/B in the relation $F = A\sqrt{x} + Bt^2$, where F is the force, x is the distance and t is time?

1) ML^2T^{-2} 2) $L^{-1/2}T^2$ 3) $L^{-1/2}T^{-1}$ 4) LT^{-2}

2. A body is projected vertically upwards. If t_1 and t_2 be the times at which it is at height h above the projection while ascending and descending respectively, then h is…

1) $\frac{1}{2}gt_1t_2$ 2) gt_1t_2 3) $2gt_1t_2$ 4) t_1t_2

3. In the device the acceleration of block A is $1\,m/s^2$. The acceleration of block B will be:



1) $1\,m/s^2$ 2) $2\,m/s^2$ 3) $4\,m/s^2$ 4) $6\,m/s^2$

4. A body is moved along a straight line by a machine delivering constant power. The distance moved by in time t is proportional to $t^{p/q}$. Then the value of pq is ____.

1) 2 2) 4 3) 6 4) 8

5. The mass per unit length of a non-uniform rod AB of length L varies as $m = m_0\frac{x^2}{L}$ where m_0 is a constant and x is the distance at any point on the rod measured from end A, the centre of mass of the rod from A is:

1) $\frac{L}{3}$ 2) $\frac{L}{4}$ 3) $\frac{2L}{3}$ 4) $\frac{3L}{4}$

6. The moment of inertia of cylinder of radius a, , mass M and height h about an axis parallel to the axis of cylinder and distant b from its centre is:

1) $\frac{1}{2}M\left(a^2+2b^2\right)$ 2) $\frac{1}{2}M\left(2a^2+b^2\right)$ 3) $\frac{1}{2}M\left(a^2+b^2\right)$ 4) $\frac{1}{2}M\left(\frac{a^2}{3}+\frac{b^2}{12}\right)$

7. In four complete revolution of the cap, the distance travelled on the pitch scale is 2 mm. If there are 50 divisions on the circular scale, then calculate the least count of screw gauge___. (in mm)

1) 0.1 2) 0.01 3) 0.001 4) 1

8. A piece of metal weight 46 g in the air. When it is immersed in the liquid of specific gravity 1.24 at 27^0C, it weighs 30 g. When the temperature of liquid is raised to 42^0C, the metal piece weighs 30.5 g. Specific gravity of the liquid at 42^0C is 1.20. The coefficient of linear expansion of the metal will be:

1) $3.316\times10^{-5}\ /^0C$ 2) $2.316\times10^{-5}\ /^0C$ 3) $4.316\times10^{-5}\ /^0C$ 4) $1.316\times10^{-5}\ /^0C$

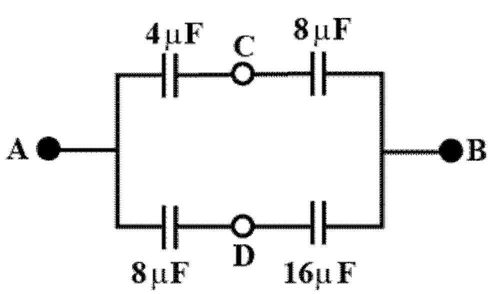
9. For a gas $\frac{R}{C_p}=\frac{2}{3}$, possibly the gas is

1) Diatomic 2) Monoatomic
3) A mixture of diatomic and monoatomic 4) Polyatomic

10. A particle moves with a simple harmonic motion in a straight line. In the first second starting from rest it travels a distance a and in the next second it travels a distance b in the same direction. The amplitude of the motion is:

1) $\frac{2a^2}{3b-a}$ 2) $\frac{3a^2}{3a-b}$ 3) $\frac{2a^2}{3a-b}$ 4) $\frac{3a^2}{3b-a}$

11. In the circuit shown, some potential difference is applied between A and B. If C is joined to D then



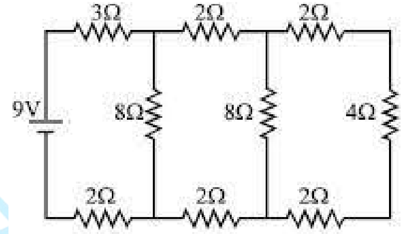
1) No charge will flow between C and D

2) Some charge will flow between C and D

3) Equivalent capacitance between A and B will change

4) Equivalent capacitance between A and B is $16\mu F$

12. In the circuit shown in the figure, the current through the 4Ω resistor is ____amp.

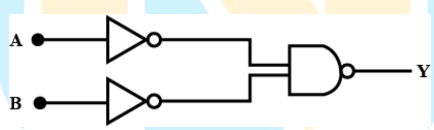


1) $\frac{1}{2}$ 2) $\frac{1}{4}$ 3) $\frac{1}{6}$ 4) $\frac{1}{8}$

13. A circuit I flows through a thin wire shaped as regular polygon of n sides which can be inscribed in a circle of radius R. The magnetic field induction at the center of polygon due to one side of the polygon is

1) $\frac{\mu_0 I}{\pi R}\left(\tan\frac{\pi}{n}\right)$ 2) $\frac{\mu_0 I}{4\pi R}\left(\tan\frac{\pi}{n}\right)$ 3) $\frac{\mu_0 I}{2\pi R}\left(\tan\frac{\pi}{n}\right)$ 4) $\frac{\mu_0 I}{2\pi R}\left(\cos\frac{\pi}{n}\right)$

14. The logic gate equivalent to the given logic circuit is



1) NAND 2) OR 3) NOR 4) AND

15. A coil has a resistance of 10 ohm and an inductance of 0.4 Henry. It is connected to an AC source of 6.5 V, $\frac{30}{\pi}Hz$. Find the average power consumed in the circuit.

1) $\frac{3}{8}W$ 2) $\frac{5}{8}W$ 3) $\frac{7}{8}W$ 4) $\frac{9}{8}W$

16. A plane electromagnetic wave of frequency 25 MHz travels in free space along the x – direction. At a particular point in the space and time, $\vec{E} = 6.3\,\hat{j}\,V/m$. What is $\vec{B}$ at this point?

1) $0.21\times10^{-8}\hat{k}T$ 2) $1.21\times10^{-8}\hat{k}T$ 3) $21.1\times10^{-8}\hat{k}T$ 4) $2.1\times10^{-8}\hat{k}T$

17. Orange light of wavelength $6000\times10^{-10}m$ illuminates a single slit of width $0.6\times10^{-4}m$. The maximum possible number of diffraction minima produced on both sides of the central maximum is ___.

1) 49 2) 99 3) 148 4) 198

18. The absorption coefficient of X – rays for a given wavelength is larges for

1) Lithium 2) Lead 3) Aluminium 4) Copper

19. Nearly 10 % of the power of a 110 *W* light bulb is converted to visible radiation. The change in average intensities of visible radiation, at a distance of 1 *m* from the bulb to a distance of 5 *m* is __. $\left(\text{in } W/m^2\right)$

1) 84×10^{-3} 2) 8.4×10^{-4} 3) 84×10^{3} 4) 84×10^{-2}

20. A nuclear explosion is designed to deliver 1 MW of heat energy, how many fission events must be required in a second to attain this power level. If this explosion is designed with a nuclear fuel consisting of uranium – 235 to run a reactor at this power level for one year, then calculate the amount of fuel needed. You can assume that the amount of energy released per fission event is 200 MeV.

1) 438.5 g 2) 384.5 g 3) 583.4 g 4) 843.5 g

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

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21. A ball is projected so as to pass a wall at a distance a from the point of projection at an angle of 45^0 and falls at a distance *b* on the other side of the wall. If *h* is the height of the wall then $h=\frac{x\,ab}{a+b}$. Then the value of *x* is ___.

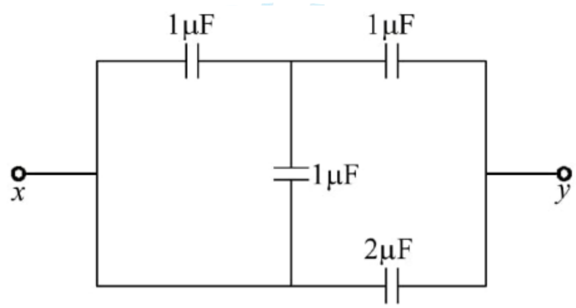
22. The gravitational force between a point like mass *M* and an infinitely long thin rod of linear mass density λ at perpendicular distance *L* from *M* is $\frac{x\,MG\lambda}{L}$. Then the value of *x* is ___.

23. A hole is made at the bottom of a large vessel open at the top. If water is filled to a height *h*, it drains out completely in time *t*. The time taken by the water column of height 2*h* to drain completely is xt then the value of x is ___.

24. A tuning fork of frequency 500 Hz is sounded and resonance is obtained at 17 cm and 52 cm of air column respectively in a resonating air column apparatus. The velocity of sound in air is ___. $\left(\text{in } ms^{-1}\right)$

25. A particle of mass m, charge - Q is constrained to move along the axis of a ring of radius a. The ring carries a uniform charge density $+\lambda$ along its circumference. Initially, the particle lies in the plane of the ring at a point where no net force acts on it. The period of oscillation of the particle when it is displaced slightly from its equilibrium is $T = x\pi\sqrt{\frac{y\,m\varepsilon_0 a^2}{\lambda Q}}$. Then the value of $(x \times y)$ is ____.

26. The equivalent capacitance between terminals x and y is $\frac{a}{6}\mu F$, then the value of a is____.



27. Force between two identical bar magnets whose centres are r metre apart is 4.8 N, when their axes are in the same line. If separation is increased to $2r$, the force between them is reduced to $\frac{x}{10}N$, then the value of x is____.

28. A ray of light is incident on the plane mirror at rest. The mirror starts turning at a uniform acceleration of $2\pi\ \text{rad/s}^2$. The reflected ray at the end of $\frac{1}{4}$ sec must have turned through an angle is θ, then the value of 4θ is ____. (in degrees)

29. Light of two different frequencies whose photons have energies 1 eV and 2.5 eV respectively, successively illuminates a metal of work function 0.5 eV. The ratio of maximum kinetic energy of the emitted electron will be $x:y$, then the value of $2x+y$ is____.

30. When $_{92}U^{235}$ undergoes fission, 0.1 % of its original mass is changed into energy. How much energy is released if 1 kg of $_{92}U^{235}$ undergoes fission is $x \times 10^{13} J$, then the value of x is____.

CHEMISTRY **MAX.MARKS: 100**

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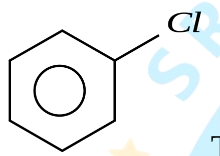
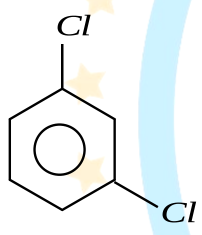
31. The weight of 2.24 liters of C_2H_6 at NTP is

A) 6.0 g B) 3.0 g C) 1.6 g D) 23 g

32. The number of waves made by an electron revolving in the fourth orbit of hydrogen atom is

A) 1 B) 4 C) 3 D) 2

33.  The dipole moment value is x debye, then the dipole moment value of

 is ___ debye.

A) $< x$ B) $= x$ C) $> x$ D) 0

34. The orbital angular momentum of 4s electron is

A) 0 B) $\frac{h}{2\pi}$ C) $\frac{2h}{\pi}$ D) $\frac{3h}{2\pi}$

35. The heats of combustion of carbon and carbon monoxide are -393.5 and –285.5 kJ mol^{-1} respectively. The heat of formation (in kJ) of carbon monoxide per mole is:

A) – 110.5 B) 110.5 C) 676.5 D) – 676.5

36. The reaction $N_2O_5(g) \rightleftharpoons 2NO_2(g) + \frac{1}{2}O_2(g)$ is started with initial pressure of $N_2O_5(g)$ is equal to 600 torr. The fraction of $N_2O_5(g)$ decomposed when the total pressure of system is 960 torr is

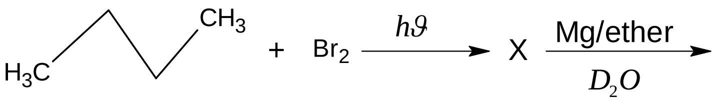
A) 0.05 B) 0.1 C) 0.2 D) 0.4

37. pH of the solution containing 50.0 mL of 0.01M HCl is

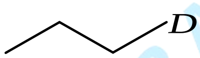
A) 4 B) 3 C) 2 D) 6

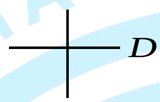
38. The hybridization of Boron in B_2H_6 is

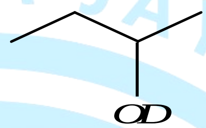
A) sp^3 B) sp^2 C) sp D) dsp^2

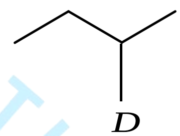
39. 

then Y is

A) 

B) 

C) 

D) 

40. The correct order for the basic nature of amines in gaseous phase is

A) $3^0 < 2^0 < 1^0$ amines B) $2^0 < 1^0 < 3^0$ amines

C) $1^0 < 2^0 < 3^0$ amines D) $1^0 < 3^0 < 2^0$ amines

41. Which one of the following reactions is used in the preparation of toluene?

A) $C_6H_6 + CH_3Cl \xrightarrow{AlCl_3}$ B) $C_6H_6 + C_2H_5Cl \xrightarrow{AlCl_3}$

C) $CH \equiv C - H \xrightarrow{Ni(CN)_2}$ D) $CH_3 - C \equiv C - H \xrightarrow[Fe\,tubes]{Red\,hot}$

42. Which one of the following is not a transition element ?

A) Sc B) Zn C) Mn D) Fe

43. Which one of the following does not undergo hydrolysis ?

A) NH_4Cl B) CH_3COONa C) $NaCl$ D) CH_3COONH_4

44. Which one of the following product is formed at cathode during electrolysis of aq.NaC*l*

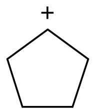
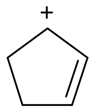
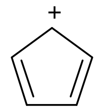
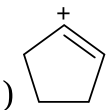
A) Na metal B) Cl_2 gas C) O_2 gas D) H_2 gas

45. For a first order reaction $A \rightarrow$ products, the concentration of A changes from 0.1M to 0.025M in 40 min. Half life of the reaction is

A) 10 mins B) 20 mins

C) 40 mins D) 100 mins

46. Among the following most stable carbocation is

A)  B)  C)  D) 

47. The product formed when glucose reacts with HI / P (red) is

A) n-hexane B) sorbitol C) gluconic acid D) sacharic acid

48. The metal ion which is precipitated when H_2S gas is passed with HCl and diluted water

A) Pb^{+2} B) Ca^{+2} C) Cu^{+2} D) Fe

49. Which one of the following complexes exhibits linkage isomerism

A) Cis $\left[CoCl_2(en)_2\right]^+$ B) Cis $\left[Co(en)_3\right]^{+3}$

C) Cis $\left[PtCl_2(NH_3)_2(py)_2\right]$ D) $\left[PtClBrINO_2\right]^{-2}$

50. In the reaction $C_2H_5OH \xrightarrow[300^oC]{Cu}$ product. The product is

A) CH_3COOH B) CH_3CHO C) C_2H_4 D) C_2H_6

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

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51. The van't Hoff factor for the complex $K_4\left[Fe(CN)_6\right]$ is (assuming 100% ionization)

52. The number of primary amines formed by the formula $C_4H_{11}N$ is (excluding stereo isomers)

53. How many of the following are amino acids Lysine, Phthalic acid, Toluidine, Tyrosine, Threonine, Arginine, Proline, Serine, Cysteine, Isoleuline, Valeric acid

54. The EAN of Ni in the complex $\left[Ni(H_2O)_6\right]SO_4$ is

55. The number of stereoisomers of the following is

$$\begin{array}{c} \quad\quad\quad\quad CH_3 \\ \quad\quad\quad\quad | \\ CH_3-CH_2-CH-CH=CH_2 \end{array}$$

56. Maximum number of ionization energy values are possible for C^{14} is

57. The bond order values of O_2, O_2^-, O_2^+ and O_2^{-2} are a, b, c & d respectively. The value of a+b+c+d is

58. The group number that has maximum number of elements in the modern periodic table is

59. The number of moles of ethyl alcohol reacts with one mole of PCl_3 is x and PCl_5 is y. Then the value of $x+y$ is

60. The number of Chlorine atoms in DDT molecule is

MATHEMATICS MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

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61. Let $A=\left\{z\in C:\left|\frac{z+1}{z-1}\right|<1\right\}$ and $B=\left\{z\in C:\arg\left(\frac{z-1}{z+1}\right)=\frac{2\pi}{3}\right\}$. Then $A\cap B$ is

1) a portion of a circle centred at $\left(0,-\frac{1}{\sqrt{3}}\right)$ that lies in the second and third quadrants only

2) a portion of a circle centred at $\left(0,-\frac{1}{\sqrt{3}}\right)$ that lies in the second quadrant only

3) an empty set

4) a portion of a circle of radius $\frac{2}{\sqrt{3}}$ that lies in the third quadrant only

62. If m_1 and m_2 be the roots of the equation $x^2+\left(\sqrt{3}+2\right)x+\sqrt{3}-1=0$ then the area of the triangle formed by the lines $y=m_1x$, $y=m_2x$ and $y=2$ is (in sq. units)

1) $\sqrt{33}-\sqrt{11}$ 2) $\sqrt{11}+\sqrt{33}$ 3) $2\sqrt{33}$ 4) 121

63. The mean of the numbers a, 8, 5, 10 is 6 and their variance is 6.8. If M is the mean deviation of the numbers about the mean, then 25 M is equal to:

1) 60 2) 55 3) 50 4) 45

64. The area enclosed by $y^2=8x$ and $y=\sqrt{2}x$ is equal to

1) $\frac{16\sqrt{2}}{6}$ 2) $\frac{11\sqrt{2}}{6}$ 3) $\frac{13\sqrt{2}}{6}$ 4) $\frac{5\sqrt{2}}{6}$

65. $\int_0^5 \cos\left(\pi\left(x-\left[\frac{x}{2}\right]\right)\right)dx$, where $[t]$ denotes greatest integer less than or equal to t, is equal to:

1) -3 2) -2 3) 2 4) 0

66. Statement-1: Period of $f(x)=\sin 3x\cos[3x]-\cos 3x\sin[3x]$, where [ ] denotes the greatest integer function, is $\frac{2\pi}{3}$

Statement-2: Period of $\{x\}$, where {} denotes the fractional part of x, is 1

A) Statement-1is True, Statement-2 is True; Statement-2 is a correctExplanation forStatement-1

B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1

C) Statement-1 is True, Statement-2 is False

D) Statement-1 is False, Statement-2 is True

67. If $\exp\left\{\left(\sin^2 x+\sin^4 x+\sin^6 x+...\text{upto}\infty\right)\ln 2\right\}$ satisfies the equation $x^2-17x+16=0$,then the value of $\frac{2\cos x}{\sin x+2\cos x}$ $\left(0<x<\pi/2\right)$ is:

1) $\frac{1}{2}$ 2) $\frac{3}{2}$ 3) $\frac{5}{2}$ 4) $\frac{7}{2}$

68. The value of $\lim_{x\to 0}\left(\left[\frac{11x}{\sin x}\right]+\left[\frac{21\sin x}{x}\right]\right)$, where $[x]$ is the greatest integer less than or equal to x ,is:

1) 32 2) 31 3) 11 4) 21

69. If $y(x)=x^x, x>0$ then, $y''(2)-2y'(2)$ is equal to

1) $8\log_e 2-2$ 2) $4(\log_e 2)^2+2$ 3) $4(\log_e 2)^2-2$ 4) $4\log_e 2+2$

70. STATEMENT – 1 : In ellipse the sum of the distances between the foci is always less than the sum of the focal distances of any point on it.
STATEMENT – 2 : The eccentricity of any ellipse is less than 1.

Statement-1is True, Statement-2 is True; Statement-2 is a correctExplanation forStatement-1

Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1

Statement-1 is True, Statement-2 is False

Statement-1 is False, Statement-2 is True

71. In a certain town, only 3 newspapers A, B & C are published. There are 1000 people living in that town and each person reads atleast one newspaper. 30 % of town population read only A. 25 % read both B & C and 10 % read all A, B & C. If 50 % read A, then number of people who read exactly two newspapers is:

1) 200 2) 250 3) 300 4) 450

72. Number of integral values of k for which the point $M(0, k)$ lies on or inside the triangle formed by the lines $y+3x+2=0$, $3y-2x-5=0$ and $4y+x-14=0$, is:

1) 2 2) 3 3) 4 4) 5

73. If A is square matrix of order 3 such that $|A|=3$, then $|adj(adj(adjA))|$ is :

1) 3^8 2) 3^9 3) 3^{10} 4) 3^{11}

74. Number of points of discontinuity of the function $f(x)=\text{sgn}(x^2-x+1)+[x^2-x+1]$ in $x\in[0, 3]$ ($[\cdot]$ denotes greatest integer function, $\text{sgn}\{\cdot\}$ denotes signum function) is

1) 7 2) 6 3) 14 4) 8

75. The least distance and the farthest distance of the point (10, 7) from the circle $x^2+y^2-4x-2y-20=0$ are 'a' and 'b' then the equation of the line passing through (a, b) and (0, 0) is

1) $y=3x$ 2) $x=3y$ 3) $x+3y=0$ 4) $3x+y=0$

76. Let $f(x)=2x^n+\lambda, \lambda\in R, n\in N$, and $f(4)=133$, $f(5)=255$. Then the sum of all the positive integer divisors of $(f(3)-f(2))$ is:

1) 61 2) 60 3) 58 4) 59

77. Let $\begin{vmatrix} 1+x & x & x^2 \\ x & 1+x & x^2 \\ x^2 & x & 1+x \end{vmatrix} = ax^5+bx^4+cx^3+dx^2+ex+f$

Column – I	Column – II
a) Value of f is	p) 0
b) Value of e is	q) 1
c) Value of $a+c$ is	r) -1
d) Value of $b+d$ is	s) 3

1) a-q,b-s,c-r,d-q
2) a-s,b-q,c-r,d-q
3) a- s,b-s,c-q,d-r
4) a-r,b-r,c-s,d-q

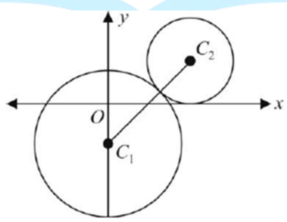
78. The distance between the lines $\bar{r} = \left(\bar{i}+2\bar{j}-4\bar{k}\right)+\lambda\left(2\bar{i}+3\bar{j}+6\bar{k}\right)$, $\bar{r} = \left(3\bar{i}+3\bar{j}-5\bar{k}\right)+\mu\left(2\bar{i}+3\bar{j}+6\bar{k}\right)$ is

1) 7 2) $\frac{\sqrt{290}}{7}$ 3) $\frac{\sqrt{293}}{7}$ 4) $\frac{\sqrt{293}}{3}$

79. Let the first term a and the common ratio r of a geometric progression be positive integers. If the sum of its squares of first three terms is 21, then the sum of first 10 terms of the given G.P:

1) 1023 2) 511 3) 2047 4) 241

80. In the given figure, the equation of the larger circle is $x^2+y^2+4y-5=0$ and the distance between centres is 4. Then the equation of the smaller circle is:



1) $\left(x-\sqrt{7}\right)^2+\left(y-1\right)^2=1$
2) $\left(x+\sqrt{7}\right)^2+\left(y-1\right)^2=1$
3) $x^2+y^2=2\sqrt{7}x+2y$
4) $\left(x+\sqrt{7}\right)^2+\left(y+1\right)^2=1$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

81. If the shortest distance between the line joining the points $(1, 2, 3)$ and $(2, 3, 4)$, and the line $\frac{x-1}{2}=\frac{y+1}{-1}=\frac{z-2}{0}$ is α, then $28\alpha^2$ is equal to____.

82. The remainder when $(2021)^{2023}$ is divided by 7 is ____.

83. There are ten boys $B_1, B_2, ..., B_{10}$ and five girls $G_1, G_2, ..., G_5$ in a class. Then the number of ways of forming a group consisting of three boys and three girls, if both B_1 and B_2 together should not be the members of group, is _____.

84. If α and β are the roots of $x^2+x+2=0$ and γ,δ are the roots of $x^2+3x+4=0$, then $(\alpha+\gamma)(\alpha+\delta)(\beta+\gamma)(\beta+\delta)$ is equal to

85. If $\sin(10^0)\sin(50^0)\sin(70^0)=\alpha$, then $16+\alpha^{-1}$ is equal to _____.

86. Let α and β be the roots of the equation $x^2+(2i-1)=0$. Then the value of $|\alpha^8+\beta^8|$ is equal to ____.

87. If the term independent of x in the expansion of $\left(\sqrt{\frac{x}{3}}+\frac{3}{2x^2}\right)^{10}$ is $\frac{m}{n}$ (m, n are relatively prime) then the value of $(m+n)$ is

88. The number of irrational terms in the expansion of $\left(5^{1/8}+2^{1/6}\right)^{100}$ is equal to

89. $50\tan\left(3\tan^{-1}\left(\frac{1}{2}\right)+2\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)\right)+4\sqrt{2}\tan\left(\frac{1}{2}\tan^{-1}\left(2\sqrt{2}\right)\right)$ is equal to ____.

90. If $A=\begin{bmatrix} a & a^2-1 & -2 \\ a+1 & 1 & a^2+4 \\ -2 & 4a & 5 \end{bmatrix}$ is symmetric, then a is